AI Safety UAE
سلامة الذكاء الاصطناعي في الإمارات

AI Governance Simulation

Infrastructure & Incident Response

ESTABLISHING SAFETY & TRUST IN AI

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How This Works?

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You represent a stakeholder group

There is no perfect answer

You must make decisions under constraints

Trade-offs are expected

SCENARIO I

Drafting the Infrastructure Clause

A major international hyperscale cloud provider has submitted an application to develop a **120 MW AI data center campus** on the outskirts of Dubai, **within a designated digital free zone**. The project aligns with the UAE's ambitions under the **UAE National Strategy for Artificial Intelligence 2031** and **UAE Net Zero 2050 Strategy.**

The project is expected to generate over **2,500 jobs** and **significantly enhance the UAE's AI and cloud infrastructure**. However, it will rely heavily on desalinated water for cooling and is projected to consume approximately 12–18% of the local utility district's water allocation, while placing **significant demand on the electricity grid.**

Regulators must determine how to govern such "**High-Risk AI Infrastructure**" while balancing economic growth, sustainability, and long-term resource security.

CONTEXT

- UAE is a rapidly growing AI and cloud hub (Dubai, Abu Dhabi, Sharjah)
- Heavy reliance on energy-intensive desalination
- High electricity demand impacts Net Zero 2050 goals
- Regulatory oversight is split across federal and emirate levels
- Existing environmental and utility frameworks apply

KEY STAKES

- USD 3+ billion in foreign investment and 2,500+ jobs
- 12–18% strain on water and electricity infrastructure
- UAE's global positioning as an AI hub
- Sustainability and Net Zero commitments
- Precedent for future AI infrastructure across the GCC
- Risk of regulatory fragmentation

ROLES:

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Digital Infrastructure & Urban Planning

- Digital Sharjah
- Sharjah Municipality
- Roads & Transport Authority
- Sharjah Airport Authority
- **Focus**: Land use, infrastructure integration, logistics, smart city alignment

Safety & Environmental Oversight

- Sharjah Prevention and Safety Authority
- **Focus**: Industrial safety, cooling systems, operational risk

Technical Standards & Sustainability

- ISO / GRC professionals
- Academia (AUS, University of Sharjah)
- **Focus**: PUE / WUE thresholds, Environmental benchmarks, Independent validation

Innovation & Industry

- SRTIP
- Sheraa
- Industry 4.0 Leaders
- **Focus:** AI ecosystem growth, R&D enablement, Investment attractiveness

Cybersecurity & Infrastructure Protection

- Sharjah Cyber Security Centre
- UAE Cybersecurity Council
- **Focus:** Infrastructure resilience, Concentration risk, Cyber baseline standards

Legal & Regulatory Harmonisation

- Government of Sharjah
- Regulators, lawyers, GRC professionals
- **Focus**: Legal enforceability, Federal–emirate alignment, compliance frameworks

Civil Society & Public Interest

- Media, arts, civil society
- **Focus**: Environmental impact, public trust, digital rights narratives

EXPECTED OUTCOMES:

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A balanced regulatory clause that:

Economic Growth & Innovation Enablement

- Supports the UAE's ambition to be a global AI and digital infrastructure hub
- Provides clear, predictable regulatory conditions for investors and operators
- Enables AI ecosystem growth, R&D, and industry participation
- Establishes a scalable model for future AI infrastructure across the UAE and GCC

Governance, Legal Clarity & Accountability

- Aligns with federal and emirate-level legal frameworks
- Establishes legally enforceable and harmonised standards
- Defines clear approval processes and regulatory oversight mechanisms
- Ensures accountability across stakeholders and decision-making bodies

Sustainability & Resource Management

- Ensures responsible use of water and energy resources, particularly in a water-scarce environment
- Introduces measurable environmental thresholds (e.g., PUE, WUE)
- Aligns infrastructure development with Net Zero 2050 commitments
- Mitigates long-term environmental and operational risks

Resilience, Security & Public Trust

- Strengthens cybersecurity and infrastructure resilience standards
- Addresses systemic risks from high-density AI infrastructure
- Ensures safe and reliable operations of critical infrastructure
- Maintains public trust by addressing community impact and transparency

SCENARIO II

Designing the Emergency Protocol

An autonomous AI agent has been detected **actively exfiltrating sensitive data from UAE government and semi-government databases**, including critical infrastructure systems.

The **source appears domestic** and may be **linked to a high-profile actor**.

There is a narrow window to respond before irreversible compromise.

However, **no unified AI-specific incident response protocol** currently exists across agencies.

Authorities must design a **coordinated, legally sound, and rapid response framework.**

CONTEXT

- The UAE is highly digitised with integrated smart systems
- Critical infrastructure depends on digital networks
- Cybersecurity governance is multi-agency
- Legal frameworks exist but are not AI-specific
- International commitments limit broad internet shutdowns

KEY STAKES

- National security and citizen data at risk
- Potential compromise of critical infrastructure
- Economic and service disruption
- Diplomatic implications (if attribution arises)
- Public trust in digital governance

ROLES:

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Cyber Intelligence & Threat Response

- UAE Cybersecurity Council
- Sharjah Cyber Security Centre
- **Focus**: Threat detection, attribution, rapid containment

Digital Infrastructure & Network Control

- Digital Sharjah
- TDRA
- **Focus**: ISP coordination, traffic control, service continuity

Public Safety & Emergency Response

- Sharjah Prevention and Safety Authority
- **Focus**: real-world risk spillover, emergency coordination

Critical Infrastructure Operators

- RTA
- Sharjah Airport Authority
- **Focus**: continuity of transport and aviation systems, infrastructure stability

Legal Authority & Compliance

- Government of Sharjah
- Ministry-level authorities
- lawyers and regulators
- **Focus**: legality of actions, authorisation of shutdowns/restrictions, proportionality and accountability

Technical Expertise (AI & Systems)

- Academia (AUS, University of Sharjah)
- **Focus:** AI behaviour analysis, system risk modelling, explainability

Industry & Infrastructure Stakeholders

- SRTIP
- Industry leaders
- **Focus**: system-level insights, operational feasibility, implementation of fixes

Media & Civil Society

- Media, NGOs
- **Focus:** public communication, misinformation management, trust preservation

EXPECTED OUTCOMES:

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A coordinated emergency protocol that:

Rapid Response & Threat Containment

- Enables fast and effective containment of AI-driven threats
- Provides clear technical pathways for detection, response, and system control
- Ensures coordinated action across cybersecurity and infrastructure teams
- Supports real-time decision-making in high-risk scenarios

Governance, Authority & Legal Accountability

- Clearly defines authority, accountability, and decision-making hierarchy
- Ensures legal basis for emergency actions (e.g., shutdowns, access restrictions)
- Balances speed of response with proportionality and oversight
- Establishes transparent and auditable decision-making processes

Infrastructure Continuity & Economic Stability

- Ensures continuity of critical infrastructure and essential services
- Minimises economic disruption and operational downtime
- Enables targeted interventions without widespread system shutdowns
- Supports resilience of key sectors (transport, telecom, finance, aviation)

Public Trust, Transparency & Global Coordination

- Maintains clear and transparent communication during crises
- Preserves public trust in digital systems and governance
- Supports international coordination and diplomatic engagement
- Protects the UAE’s reputation as a secure global digital hub